Robotic AI Memory: A Focused Literature Map

> **Search cutoff:** 12 September 2026
> **Primary window:** 2024–2026, with older work included only when it defines a necessary baseline
> **Focus:** efficient memory for object-rich 3D environments, especially the interface between robotic world models and transformer memory
> **Status labels: Published/accepted** means a venue is confirmed by the proceedings or paper page; **Preprint** means the result should be treated as provisional.

Executive conclusion

The field is converging on **multi-timescale hybrid memory**, but its pieces remain separated:

1. **Explicit 3D/4D world memory** provides persistent object identity, geometry, time, and interpretable updates.
2. **Retrieval memory** selects the small part of a long history needed for the current question or action.
3. **Transformer-native memory** turns selected history into tokens, keys/values, recurrent state, or fast weights that a policy can consume efficiently.
4. **Episodic/task memory** records action progress, success, failure, and semantic events across long tasks.

The cutting edge is therefore not one universally best memory module. It is an architecture that assigns different jobs to different stores. Recent systems already combine short- and long-term memory, KV caching and learned memory, or explicit 3D mapping and semantic retrieval. Merely proposing "KV cache plus neural memory" would no longer be a strong novelty claim [4, 10–13, 29–36].

The most defensible remaining gap is narrower:

> ***Under a fixed token and latency budget, how should an embodied transformer retrieve a small, fresh, object-grounded working memory from a persistent 3D/4D environment memory?***

This is directly about both robotics AI and transformers. It does not require building a complete robot. It isolates the interface between an editable robotic world model and the bounded context used by a transformer policy or reasoner.

The 15 papers to read first

Priority	Ref.	Paper and status	Why it matters for this topic	Important limitation
1	[1]	**SuperMap** – RSS 2026	Strong current systems baseline for a queryable 4D scene graph with persistent identities, confidence updates, reactivation, and stale-object pruning.	Navigation-oriented; the map is not a transformer-native working memory.
2	[2]	**HitMem** – ECCV 2026	Most direct new paper on hierarchical object memory, temporal decay, and retrieval after objects are displaced.	New benchmark and primarily simulated evaluation; no end-to-end manipulation policy.
3	[3]	**DynaMem** – preprint, 2024/2025	Direct evidence that dynamic spatio-semantic memory improves real mobile pick-and-drop when objects move, appear, or disappear.	Point-cloud memory and system heuristics, not a learned transformer memory.

Priority	Ref.	Paper and status	Why it matters for this topic	Important limitation
4	[4]	**MEM** – preprint, 2026	High-profile robot-policy result combining video short-term memory with language long-term memory for tasks lasting up to 15 minutes.	Remembers events and task state rather than a persistent metric object map.
5	[5]	**3DLLM-Mem** – NeurIPS 2025	Clear working-memory/episodic-memory formulation; working tokens query spatial and temporal history. Introduces the large 3DMem-Bench.	Focused on reasoning and action prediction in datasets, not online editable world state.
6	[6]	**HALO** – RSS 2026	Shows that retrieval itself must be learned and supervised; sparse top-K attention reduces drift in long-horizon visuomotor control.	History horizon is minutes, but memory lacks explicit metric/object structure.
7	[7]	**JanusVLN** – ICLR 2026	The cleanest direct bridge to KV caching: separate fixed-size spatial-geometric and visual-semantic KV memories.	Navigation only; fixed windows can forget rare but important objects.
8	[8]	**Planning with an Embodied Learnable Memory** – ICLR 2026	A VLM learns to add, remove, and update an environment representation for planning.	Textual memory is editable but discards precise geometry and object evidence.
9	[9]	**NativeMEM** – preprint, 2026	Compresses each historical frame-view to one token with the VLA's native vision encoder, giving long history with low overhead.	A frame token is not an independently editable object record; preprint and code maturity should be checked.
10	[10]	**LaMem-VLA** – preprint, 2026	Curates, retrieves, condenses, and weaves short- and long-term latent memory tokens directly into VLA reasoning.	Simulation evidence; no metric 3D grounding or persistent object identity.
11	[11]	**LoGeR** – preprint, 2026	Important novelty boundary: already combines non-parametric sliding-window attention with parametric TTT memory.	Solves geometric reconstruction, not semantic object memory or robot decisions.
12	[12]	**STAC** – CVPR 2026	Separates a short-term temporal cache from a long-term voxel-aligned spatial cache; directly studies compression and speed.	Reconstruction cache, not open-vocabulary object or action memory.
13	[13]	**SpatialMem** – preprint, 2026	Hierarchical 3D anchors and object nodes link coordinates, evidence crops, embeddings, and descriptions for compact retrieval.	Only three evaluated scenes and an offline video-memory setting.
14	[14]	**R4DSG** – ACM Multimedia 2026	Uses stable anchors and relative changes to preserve object identity without requiring a globally aligned dense map.	Small object-QA subset, offline construction, and wearable video rather than robot control.
15	[15]	**DovSG** – RA-L 2025	Dynamic open-vocabulary 3D scene graph with local updates for long-term mobile manipulation.	The update logic is structured/system-level rather than transformer-native learning.

Scope and search protocol

Included

- Memory that persists across observations, actions, rooms, or episodes.

- Object-centric, metric-semantic, scene-graph, 3D/4D, episodic, retrieval, KV-cache, recurrent-state, and fast-weight memories.
- Navigation, embodied question answering, mobile manipulation, and VLA control when the memory mechanism is reusable.
- Papers with evidence about write, update, retrieval, compression, forgetting, or action conditioning.

Excluded from the core map

- Ordinary SLAM without semantic or temporal memory.
- Generic long-context language models with no plausible embodied-memory transfer, except Titans as the relevant neural-memory baseline [39].
- Pure action representation, grasping, or world-model papers that do not study memory.
- Video QA papers that retrieve clips but do not preserve spatial or object state.
- Additional streaming-3D variants unless they introduce a distinct memory mechanism.

Search method

The search began with the supplied memory report, representation report, and June 2026 literature map. Their references and claims were then checked against primary paper pages and proceedings. Forward searching targeted combinations of:

- `robot memory`, `embodied memory`, `spatial memory`, `3D memory`, and `4D scene graph`;
- `dynamic object`, `object relocation`, `open-vocabulary`, `mobile manipulation`, and `long horizon`;
- `VLA memory`, `visuomotor retrieval`, `episodic memory`, and `working memory`;
- `KV cache`, `cache compression`, `test-time training`, `fast weights`, `neural memory`, and `streaming 3D`.

The map favors 2025–2026 work, but retains 2024 systems that remain the direct baselines for dynamic object memory. Venue status was separated from preprints because several of the most recent claims have not yet received peer review.

A functional taxonomy: what operation does the memory solve?

This taxonomy is more useful than grouping papers only by application.

Operation	Core question	Strong examples	What remains weak
Write	What deserves to enter memory?	HALO's task-relevant supervision [6]; MEM's multi-modal abstraction [4]; LaMem-VLA's curator [10]	Most 3D maps write every confident observation, not action consequences or task value.
Associate	Is this observation the same object/event as before?	SuperMap [1], R4DSG [14], ConceptGraphs [18]	Open-world identity remains brittle under occlusion, relocation, duplicates, and viewpoint change.
Represent	In what format should memory live?	4D graphs [1, 14, 16], hierarchical object memories [2, 13], latent tokens [9, 10], KV states [7], fast weights [11, 32, 39]	No representation is simultaneously metrically precise, semantically rich, editable, compact, and native to a policy.
Update	How does new evidence revise an old belief?	Confidence and pruning in SuperMap [1]; local graph updates in DovSG [15]; dynamic point clouds in DynaMem [3]; learned textual edits in EPM [8]	Action-conditioned updates with calibrated uncertainty are still uncommon.
Retrieve	Which small part of memory should enter the current computation?	HALO [6], 3DLLM-Mem [5], HitMem [2], LaMem-VLA [10], EmbodiedRAG [24]	Retrieval is often evaluated only by downstream success, obscuring stale-object and wrong-evidence failures.
Compress/consolidate	How can history remain bounded?	NativeMEM [9], STAC [12], AstraNav-Memory [27], LoGeR [11], HAMLET [30]	Compression can erase object identity, geometry, or rare events; few papers measure what was forgotten.
Forget/invalidate	When is old information harmful?	SuperMap's stale-content pruning [1], HitMem's temporal decay [2], ELMUR's replacement/blending [31]	Forgetting policies rarely distinguish "unseen" from "disproved," or reversible uncertainty from deletion.

Operation	Core question	Strong examples	What remains weak
Read for action	How does memory change the policy?	MEM [4], HALO [6], NativeMEM [9], LaMem-VLA [10], MemoryVLA [29], HAMLET [30]	The policy-side literature rarely exposes a queryable, persistent 3D object state.

Literature cluster A: explicit environmental memory

A1. Dynamic 3D/4D object and scene-graph memory

This cluster is closest to the user's motivating problem: an agent must remember what objects exist, where they were, how certain that belief is, and how the scene changed.

- **SuperMap [1]** is the strongest deployable reference architecture. It runs geometric SLAM frequently and open-vocabulary perception asynchronously, then maintains persistent object identities through 3D association and reactivation. Its important contribution for a memory proposal is not simply the scene graph; it is the explicit treatment of confidence, occlusion, relocation, and stale content.
- **HitMem [2]** turns dynamic retrieval into the central problem. A lightweight support-relation graph handles coarse search, detailed geometry is retained only where useful, temporal decay lowers trust in stale entries, and displacement triggers a specialized two-stage search. It is unusually close to the proposed retrieval question.
- **DynaMem [3]** is the most direct mobile-manipulation proof point. Its robot searches beyond stale memory and updates a 3D spatio-semantic point-cloud store as objects move, appear, or disappear. The reported average pick-and-drop success is 70% for non-stationary objects, more than twice the static-system baseline in its evaluation.
- **DovSG [15]** represents the environment as an open-vocabulary object graph and supports local graph edits after interactions, avoiding full reconstruction. It demonstrates that edit locality matters for long-term operation.
- **DAAAM [16]** builds a globally consistent hierarchical 4D scene graph with rich localized descriptions and exposes it to a tool-calling agent. It improves semantic richness without abandoning metric-temporal structure, but remains a preprint.
- **Khronos [17]** is the useful temporal-SLAM foundation: a fast active window is reconciled with a slower long-term metric-semantic model. It established the multi-timescale pattern now reappearing in transformer memories.
- **ConceptGraphs [18]** is the open-vocabulary object-graph baseline. It showed that a compact graph of object nodes and relations can support language querying and planning, though it is fundamentally static compared with the papers above.

Takeaway: explicit maps are good long-term sources of truth, but they are not automatically efficient working memories. They need a retrieval and serialization layer before a transformer can use them under a fixed context budget.

A2. Relative and hierarchical object memory

- **SpatialMem [13]** treats metric 3D as an indexing scaffold. Structural anchors support object nodes, while each object connects coordinates to visual evidence, embeddings, and descriptions. This is a practical design for explainable retrieval, but its evidence base is still small.
- **R4DSG [14]** avoids requiring globally precise reconstruction. Stable anchors and anchor-relative transitions preserve persistent object state in long egocentric video. This suggests a useful fallback when robot localization is imperfect.
- **Open3DSG [19]** and **HOV-SG [20]** supply complementary representation baselines: queryable open-vocabulary relations and a floor–room–object hierarchy. Their main value here is structural indexing rather than temporal memory.

Design implication: the environment memory should support both absolute metric fields and relative relations. Absolute coordinates help planning; support and anchor relations remain useful when pose estimates drift.

Literature cluster B: retrieval and reasoning over long embodied history

- **3DLLM-Mem [5]** separates current working-memory tokens from stored episodic spatial and temporal features. The current tokens query only useful history. Its accompanying 3DMem-Bench contains more than 26,000 trajectories and 2,892 embodied tasks, making it one of the best evaluation resources for selective memory.

- **HALO [6]** demonstrates that attention does not automatically learn good retrieval. VLM-generated question-answer supervision teaches task relevance, while sparse attention limits retrieval to the most relevant history and reduces closed-loop drift. This is a key methodological lesson for any proposed retriever.
- **Planning with an Embodied Learnable Memory [8]** learns memory edits inside a VLM: additions, removals, and updates are optimized for downstream planning. It is compelling evidence that the write/update policy itself can be learned, although the textual state is geometrically weak.
- **GraphEQA [21]** combines a real-time metric-semantic graph with task-relevant images. It is a good multimodal read interface: the structured graph narrows the problem while raw visual evidence lets the reasoner verify details.
- **Enter the Mind Palace [22]** studies active embodied QA across past, present, and future. The agent decides when to consult memory, explore, or stop, shifting attention from passive recall to the value of information.
- **3D-Mem [23]** uses memory and frontier snapshots for embodied exploration and reasoning. Its explicit criticism of oversimplified scene graphs motivates retaining visual evidence, not only symbolic triples.
- **EmbodiedRAG [24]** retrieves a task-relevant 3D subgraph for an LLM planner, reporting order-of-magnitude token reduction and substantially lower planning time. It is the clearest early baseline for "world model first, bounded transformer context second."
- **ReMEmbR [25]** stores captioned observations with time and pose in a vector database for long-horizon robot navigation QA. It is easy to implement, but similarity search over captions can confuse duplicated objects and stale observations.
- **Embodied VideoAgent [26]** updates persistent memory from egocentric video and embodied sensors when object-related activities are perceived. It shows the value of event-sensitive writes.
- **MemoryExplorer / LMEE-Bench [28]** uses reinforcement learning to train an MLLM to query memory actively during embodied exploration. It is useful when retrieval decisions should optimize future information gain rather than only current similarity.

Takeaway: retrieval must be treated as a learned or explicitly evaluated policy. The core independent variable should be *which items enter working context under the same budget*, not simply whether a system "has memory."

Literature cluster C: memory inside robot policies and VLAs

This cluster is the strongest link to the transformer course.

- **MEM [4]** divides memory by abstraction and timescale: video features preserve recent perceptual detail, while language summarizes long-term semantic progress. Its real-world tasks last up to 15 minutes. The architecture is strong for task progress, but it does not offer an object-level 3D source of truth.
- **NativeMEM [9]** reuses the pretrained VLA's vision encoder to compress every historical camera frame to a single token. This avoids a separate planner or fresh memory model and keeps latency low. It is a compelling efficiency baseline for any proposed object-token alternative.
- **LaMem-VLA [10]** makes memory context-native. A curator writes dual short/long stores, a seeker retrieves, a condenser produces fixed-size latent tokens, and a weaver inserts them into the same sequence as current visual, language, and action tokens. This is very close to the desired transformer interface, but the stored experience is not spatially grounded.
- **MemoryVLA [29]** combines a perceptual-cognitive memory bank, a working-memory query, redundancy management, and a diffusion action expert. It confirms the value of separating working and longer-term memory.
- **HAMLET [30]** summarizes each timestep into moment tokens, then consolidates them with a lightweight memory module. It is a practical retrofit to pretrained VLAs and reports large gains on history-dependent real-world tasks.
- **SAM2Act+ [37]** adds a memory bank, encoder, and attention mechanism to a multi-view robot transformer and introduces MemoryBench for spatial memory and action recall. It is an important manipulation baseline, though the history is relatively short and task-specific.
- **ELMUR [31]** adds memory embeddings and layer-wise read/write operations to a transformer, with fixed-capacity replacement or blending. Its results include long-horizon partially observable environments and robot manipulation, but the memory does not expose objects or geometry.

- **HyMeS [40]** lets a high-level agent manage semantic history and verify subtask completion while a Markovian VLA executes low-level behavior. This is useful for system decomposition, but it is less suitable if the paper's contribution must lie inside the transformer memory itself.

Takeaway: VLA memory is rapidly becoming crowded. A new proposal should not be "add a memory bank to a VLA." Its differentiator should be the *content and retrieval policy*: object-grounded, temporally valid, metrically useful, and budget-controlled.

Literature cluster D: KV caches, recurrent state, and neural fast-weight memory for 3D

D1. KV/cache memory

- **JanusVLN [7]** constructs separate spatial-geometric and visual-semantic memories directly from historical KV caches and retains an initial plus sliding window. It proves that compact KV state can encode complementary embodied information.
- **STAC [12]** uses a decayed temporal cache for short-term detail and a voxel-aligned spatial cache for long-term context. This is a particularly relevant source for token selection by both time and space.
- **AstraNav-Memory [27]** compresses image-centric navigation memory to roughly 30 tokens per image and reports 10–20× token compression, showing how aggressively perception history can be reduced.
- **InfiniteVGGT [33]** adaptively prunes KV state and retains an anchor frame for long streaming 3D reconstruction. It is useful as a budgeted-cache baseline, not as the proposal itself.
- **RnG [34]** treats KV activations as an implicit 3D representation that can be queried by a target camera pose. It shows that cached transformer state can be spatially addressable, even without explicit objects.

D2. Recurrent and neural/parametric memory

- **LoGeR [11]** already combines a non-parametric local attention window with a parametric test-time-trained global memory. This invalidates a broad novelty claim around "combining KV cache with neural memory."
- **Scal3R [32]** inserts lightweight adaptive memory units into a feed-forward geometry model and updates them at test time. It is the closest neural-memory reference for persistent 3D global context.
- **tttLRM [35]** compresses observations into fast weights at test time and decodes an implicit 3D representation. It demonstrates parameter memory as a scene representation.
- **Mem3R [36]** is another hybrid boundary paper: an implicit fast-weight MLP supports tracking while an explicit fixed-size token state supports geometric mapping.
- **CUT3R [38]** maintains a persistent recurrent 3D state across observations. It is a strong continuous-state baseline but difficult to edit or query by object identity.
- **Point4Cast [41]** keeps an evolving latent spacetime representation that can answer for past, current, and future geometry. It broadens memory from retention to forecasting.
- **Titans [39]** is the general transformer mechanism: attention acts as short-term memory while a neural module learns long-term associations through test-time updates. Its value is conceptual transfer; it is not evidence that neural parameter memory alone is suitable for robotic world state.

Takeaway: parametric memory offers bounded capacity and learned consolidation; explicit memory offers auditability and exact edits. The unresolved question is not which one should replace the other, but how an explicit object store should select and encode a bounded neural working context.

Benchmark map

Benchmark/resource	Ref.	What it tests	Best use in a feasible proposal	Main caveat
Dyna-THOR	[2]	Object relocation in dynamic simulated scenes	Staleness, relocation, and search-cost evaluation	New benchmark tied closely to HitMem.
3DMem-Bench	[5]	Long spatial-temporal QA, captioning, planning/action tasks	Broad offline evaluation of retrieval under long histories	Full model training may be costly.
MemoryBench	[37]	Manipulation tasks requiring spatial memory and action recall	Policy-level evidence if compute permits	Narrow simulated task distribution.
OpenEQA	[42]	Episodic and active embodied QA in real scans/environments	Standard spatial reasoning and evidence retrieval	QA accuracy is not manipulation success.
LMEE-Bench	[28]	Multi-goal navigation plus memory-based QA	Active retrieval and exploration	Focuses on navigation rather than object manipulation.
NaVQA / OC-NaVQA	[16, 25]	Time- and location-grounded robot QA	Temporal and geometric memory queries	Evaluation depends on VLM/LLM judge behavior.
EgoLifeQA object subset	[14]	Where/when/why questions over long egocentric video	Relative object-state memory without perfect poses	Small subset and offline construction.
OpenLex3D	[43]	Open-vocabulary 3D segmentation and object retrieval across 23 scenes	Object-centric versus dense retrieval analysis	Static datasets; no actions or temporal change.

What is already solved—and what is not

Claims that are no longer novel

- **"Use a scene graph as robot memory."** ConceptGraphs, DovSG, DAAAM, SuperMap, HitMem, and R4DSG already cover static, dynamic, hierarchical, and 4D variants [1–3, 14–18].
- **"Use attention to retrieve history."** 3DLLM-Mem, HALO, MemoryVLA, LaMem-VLA, and ELMUR all use learned attention or query-based memory access [5, 6, 10, 29, 31].
- **"Compress old frames into tokens."** NativeMEM, HAMLET, AstraNav-Memory, and LaMem-VLA make this a crowded line [9, 10, 27, 30].
- **"Combine short- and long-term memory."** MEM, LaMem-VLA, MemoryVLA, JanusVLN, STAC, Khronos, and LoGeR already instantiate this idea in different forms [4, 7, 10–12, 17, 29].
- **"Combine KV/attention memory with neural fast-weight memory."** LoGeR is the direct counterexample, with Mem3R and Scal3R occupying adjacent territory [11, 32, 36].
- **"Make memory action-conditioned."** This remains underdeveloped but cannot be claimed as untouched: VLA memories store action or task history, DovSG/DynaMem update after interactions, and EPM learns state edits [3, 8–10, 15, 29, 30].

The integration gap that remains defensible

No examined peer-reviewed system demonstrates all of the following together:

1. persistent open-vocabulary object identity in a changing 3D environment;
2. explicit temporal validity, uncertainty, and reversible invalidation;
3. task-conditioned retrieval under a fixed token and latency budget;
4. native consumption by a transformer/VLA rather than a separate symbolic planner;
5. evaluation that separates retrieval quality, stale-memory error, efficiency, and downstream task success.

This is an **interface and evaluation gap**, not an invitation to build every component from scratch.

What problems have not been solved?

The papers solve important pieces of robotic memory, but the following problems remain open or only partially addressed. They are ordered by relevance to an object-rich, long-lived robot rather than by theoretical difficulty.

Priority	Unsolved problem	Why current methods are insufficient	What a convincing solution must demonstrate
1	**Persistent object identity under change**	Scene graphs and 3D memories can associate repeated observations, but identity remains brittle when multiple objects look alike, an object is hidden for a long period, localization drifts, or another agent relocates it [1–3, 14–18].	Maintain object identity across occlusion, relocation, duplicate instances, and re-entry; report identity switches and recovery, not only task success.
2	**Temporal validity and belief revision**	Temporal decay and stale-content pruning exist [1, 2], but most memories do not clearly distinguish "not recently observed," "probably moved," "confirmed absent," and "superseded by newer evidence."	Represent belief state and evidence time explicitly; revise or reactivate beliefs without irreversibly deleting useful history.
3	**Action-conditioned causal memory writes**	VLA memories record task history, while DynaMem, DovSG, and EPM update after interaction [3, 8–10, 15, 29, 30]. Few systems explicitly bind *before-state, commanded action, observed after-state, outcome, and confidence* into one object-level event.	Show that action-linked writes improve prediction of object state, failure recovery, or future action selection over observation-only writes.
4	**Retrieval under a hard resource budget**	Learned attention, semantic search, and graph filters retrieve useful history [2, 5, 6, 10, 24], but comparisons rarely hold tokens, bytes, and latency constant.	Compare retrieval policies at equal budgets and report the accuracy–latency–memory Pareto frontier.
5	**Alignment between explicit world state and latent transformer memory**	Explicit 3D/4D maps are editable and auditable; latent tokens, KV caches, recurrent state, and fast weights are efficient for neural computation [1, 7, 9–13, 29–36]. There is no standard interface that keeps the two synchronized.	Define how object records become tokens/KVs, how token use maps back to evidence, and how updates propagate without contradictory states.
6	**Selective consolidation and forgetting without losing rare events**	Fixed windows, pruning, compression, replacement, and fast weights bound memory growth [7, 9, 11, 12, 27, 31–36, 39], but compression is usually judged by downstream averages rather than the exact information lost.	Preserve rare task-critical events while removing redundancy; measure deletion errors and recovery from premature forgetting.
7	**Calibrated uncertainty and evidence provenance**	Confidence appears in mapping systems [1, 2], and some multimodal memories retain images [13, 21, 23], but transformer answers/actions are seldom tied to inspectable evidence with calibrated uncertainty.	Return evidence pointers and uncertainty with each retrieved fact; abstain or explore when evidence is stale or contradictory.
8	**Active memory verification**	Active EQA and MemoryExplorer decide when to explore or consult memory [22, 28], but most systems retrieve passively even when a short observation could resolve uncertainty.	Learn when to trust memory, when to query another store, and when physical re-observation is worth its navigation cost.
9	**Multi-session and lifelong robustness**	Many evaluations cover one episode, a short deployment, or an offline video. Even "long-horizon" policy memories typically span minutes rather than weeks of environmental changes [4, 6, 9, 10, 26, 29, 30].	Evaluate bounded memory across repeated sessions, map drift, changing object inventories, and software/model updates.

Priority	Unsolved problem	Why current methods are insufficient	What a convincing solution must demonstrate
10	**Diagnostic evaluation across the memory pipeline**	Existing benchmarks emphasize final QA, navigation, or manipulation success [2, 5, 28, 37, 42, 43]. A failure can originate in perception, association, writing, retrieval, reasoning, or control.	Report component-level metrics so a gain can be attributed to memory rather than a stronger detector, language model, or policy.

The central unresolved tension

The field has not found a representation that is simultaneously:

- **precise enough** for metric planning and manipulation;
- **semantic enough** for open-vocabulary requests;
- **temporally valid** in a changing environment;
- **compact enough** for bounded onboard computation;
- **editable and auditable** at the level of individual objects; and
- **native enough** to participate directly in transformer attention.

Dense geometry, scene graphs, episodic databases, latent tokens, KV caches, and fast weights each satisfy only part of this list. The research opportunity lies in designing and evaluating the interface between them, not declaring one representation universally superior.

Potential points of innovation

1. Object-indexed transformer working memory — strongest recommendation

Use a persistent 3D/4D object-event store as long-term memory, but materialize only the top K task-relevant records as transformer memory tokens. Retrieval would combine semantic similarity, spatial constraints, temporal validity, uncertainty, and expected task value.

Innovation claim: existing map memories are not transformer-native, while existing transformer/VLA memories do not preserve an editable metric object state [1–10, 13–18, 29–31]. The contribution is the budgeted object-to-token interface and its evaluation.

2. Versioned object memory with reversible invalidation

Represent each object as a sequence of time-stamped hypotheses rather than one overwritten location. States such as `active`, `stale`, `displaced`, `confirmed absent`, and `reactivated` would retain evidence and uncertainty.

Innovation claim: extend temporal decay and pruning [1, 2] into explicit belief revision. This would directly address confidently retrieving an object's obsolete location.

3. Action-conditioned object-event writes

Store a compact transition record:

```
object before–state + robot action + after–observation
                    + success/failure + confidence
```

The write gate would decide whether the transition changes persistent world state, belongs only in short-term episodic memory, or should be ignored as redundant.

Innovation claim: connect the action history captured by VLA memories [4, 9, 10, 29, 30] to the explicit object updates of DynaMem, DovSG, and EPM [3, 8, 15].

4. Retrieval trained with relevance, freshness, and cost objectives

Train or tune the retriever with a multi-objective loss rather than downstream answer loss alone:

```
L = L_task + λ1 L_evidence + λ2 L_staleness + λ3 L_budget
```

Here, `L_evidence` rewards retrieval of the ground-truth supporting object/event, `L_staleness` penalizes superseded evidence, and `L_budget` penalizes tokens or latency.

Innovation claim: extend HALO's task-relevant retrieval supervision [6] to structured spatial-temporal evidence and explicit efficiency constraints.

5. Dual-address retrieval: semantic plus spatial/topological

Give every record two complementary addresses: a learned semantic key and a structured location key such as room, support object, neighborhood, or relative anchor. Queries can first filter structurally and then rank semantically—or reverse the order depending on the question.

Innovation claim: combine the efficient hierarchy of HitMem and SpatialMem [2, 13] with transformer-native semantic retrieval [5, 6, 10]. This is especially valuable for visually similar objects and imperfect global localization.

6. Query-conditioned KV materialization

Instead of preserving all historical KVs, regenerate a small working KV cache from retrieved object/event records for each task or decision phase. The persistent database remains model-independent; the generated cache is temporary and layer-compatible.

Innovation claim: invert the usual cache-compression problem. JanusVLN, STAC, and InfiniteVGGT prune activations already in the cache [7, 12, 33]; this approach first retrieves world-state evidence, then constructs only the cache the current query needs.

7. Evidence-backed memory with calibrated abstention

Attach source frames, poses, timestamps, and confidence to each object record. If retrieved claims conflict or fall below a freshness threshold, the system should abstain, request re-observation, or launch active search.

Innovation claim: join the visual evidence retention of GraphEQA and 3D-Mem [21, 23] with uncertainty-aware temporal mapping [1, 2] and active information gathering [22, 28].

8. A memory-specific evaluation protocol

Create controlled dynamic-object episodes and report separate scores for:

- object identity retention;
- latest-state accuracy;
- stale-memory error;
- evidence Recall@K;
- action-conditioned update accuracy;
- recovery after contradiction;
- retrieval latency and token/byte footprint; and
- downstream QA, planning, or manipulation success.

Innovation claim: make the memory mechanism diagnosable. This can be a meaningful contribution even with existing models and datasets because current benchmarks rarely isolate the entire write–update–retrieve–forget chain [2, 5, 28, 37, 42, 43].

Innovation opportunities ranked for a one-semester paper

Rank	Innovation point	Research value	Implementation burden	Main risk
1	Object-indexed, staleness-aware retrieval into a fixed token budget	High	Moderate	Requires reliable object/event annotations or preprocessing.
2	Versioned object memory with reversible invalidation	High and sharply scoped	Low–moderate	Could look heuristic unless evaluated against learned and decay baselines.
3	Multi-objective retrieval with explicit evidence and compute cost	High	Moderate	Ground-truth evidence labels may need to be derived.
4	Dual semantic and spatial/topological addressing	Moderate–high	Moderate	Gains depend on query diversity and duplicated objects.
5	Memory diagnostic benchmark/protocol	High practical value	Moderate	Must show that existing benchmarks genuinely miss the measured failures.
6	Action-conditioned object-event writes	High	Moderate–high	Causal attribution is difficult in offline trajectories.
7	Query-conditioned KV materialization	High architectural novelty	High	Layer-specific KV generation may require substantial model engineering.
8	Full explicit-map + VLA + neural-fast-weight system	Potentially high	Very high	Too many coupled components for a clean one-semester contribution.

The best balance is to combine points **1, 2, and 3**: retrieve versioned object records into a fixed transformer context, explicitly penalize stale evidence, and measure both task accuracy and resource cost. Points 4–7 can become ablations or follow-on work rather than mandatory system components.

Recommended narrow proposal angle

Working title

Object-Indexed Working Memory for Embodied Transformers Under a Fixed Context Budget

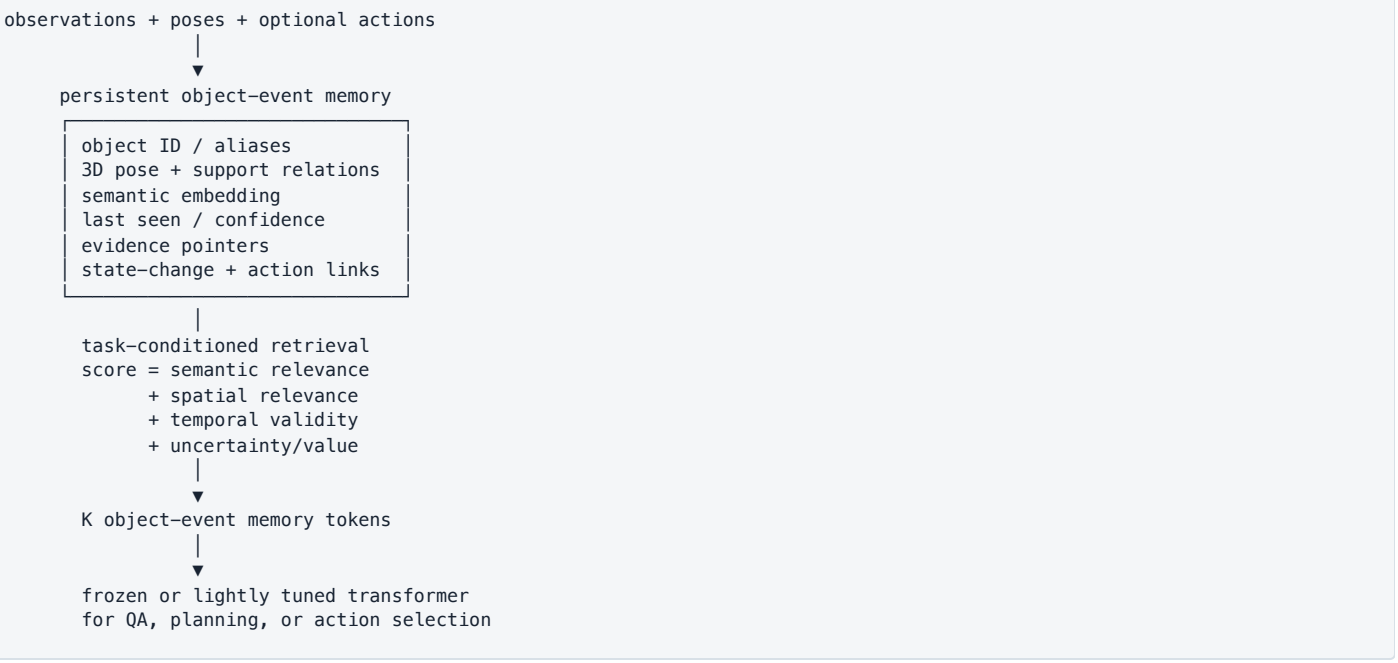
Primary research question

> *Under the same token budget, does object-, space-, and staleness-aware retrieval from an explicit 3D/4D scene memory give an embodied transformer more accurate and efficient working context than recency-only, embedding-only, or attention-only memory selection?*

Why this is a good course project

- **Robotics AI:** the stored items have object identity, 3D location/relations, time, uncertainty, and change state.
- **Transformers:** the independent variable is how retrieved memory becomes a bounded token/KV context for attention.
- **Narrow:** it studies the memory interface and retrieval policy, not perception, SLAM, grasping, and control simultaneously.
- **Measurable:** retrieval accuracy, stale-memory errors, answer/action accuracy, latency, and token count can be evaluated separately.
- **Feasible:** existing trajectories, detections, maps, and embeddings can be used; a real robot and new foundation-model training are optional.

Proposed architecture



The explicit store is the **long-term source of truth**. The selected K items are the **working memory**. A KV cache may preserve the current interaction, while an optional learned compressor converts each retrieved object/event record into one or a few native tokens. A Titans-style fast-weight module is optional and should be positioned as a learned prior or consolidation mechanism—not as the authoritative location of individual objects.

Minimal experimental design

1. Choose one dataset with dynamic object state: Dyna-THOR [2] is the cleanest; 3DMem-Bench [5] is the broader fallback.
2. Use provided object tracks/poses when available. Do not make perception accuracy the main contribution.
3. Construct a common object-event store with object ID, semantic embedding, location/relations, timestamp, confidence, and evidence pointer.
4. Hold the memory budget fixed at K records or B tokens.
5. Compare four selectors:
    - FIFO/most recent;
    - semantic nearest neighbors;
    - learned attention over a flat history;
    - proposed semantic + spatial + staleness-aware object retrieval.
6. Serialize the selected items into the same transformer input format for all conditions.
7. Evaluate:
    - Recall@K of the evidence needed for the query/action;
    - spatial-temporal answer or action accuracy;
    - stale-object error rate;
    - retrieval and end-to-end latency;
    - tokens and bytes retained;
    - calibration or abstention when memory is uncertain.

Strongest ablations

- remove object identity and retrieve frames only;

- remove spatial relations;
- remove temporal decay/invalidation;
- remove action links;
- replace structured records with the same number of learned latent tokens;
- change K to produce an accuracy–latency–memory frontier.

Hypotheses

- **H1:** structured object retrieval will outperform recency and flat semantic retrieval most strongly when many visually similar objects exist.
- **H2:** temporal validity will reduce confidently wrong answers about moved or removed objects.
- **H3:** latent/frame-token memory will be competitive on recent visual details, while object records will dominate long-range spatial and state-change queries.
- **H4:** a hybrid will form a better Pareto frontier than either a full explicit graph dump or an unstructured long history.

Alternative angles, ranked

Rank	Angle	Novelty	Delivery risk	Recommendation
1	**Budgeted object-to-token retrieval**	High enough; clearly bounded by recent work	Moderate	Best overall.
2	**Staleness-aware object memory**: distinguish unseen, uncertain, moved, removed	Strong and very focused	Low–moderate	Best if the paper must be smaller.
3	**Action-conditioned memory writes**: store before/action/after/outcome/confidence	Interesting, but adjacent VLA work is growing rapidly	Moderate–high	Good only with a carefully chosen offline dataset.
4	**Explicit graph + latent-token hybrid**	Timely but broad	High	Use as architecture framing, not the whole contribution.
5	**KV cache + neural fast weights for 3D memory**	Crowded after LoGeR/Scal3R/Mem3R	High	Do not use as the central novelty claim.

Recommended reading sequence

Pass 1 — understand the gap

Read [1], [2], [4], [6], [7], and [10]. Together they show explicit world memory, dynamic retrieval, multi-scale robot memory, supervised attention retrieval, KV memory, and transformer-native latent memory.

Pass 2 — establish baselines

Read [3], [5], [8], [9], [11], [12], and [15]. These prevent weak novelty claims and suggest feasible baselines.

Pass 3 — design the evaluation

Read [14], [23], [24], [28], [37], [42], and [43]. These define the benchmark and evidence-retrieval choices.

Pass 4 — optional neural-memory extension

Read [31], [32], [35], [36], and [39]. Only add fast weights or layer memory if the core object-to-token retrieval experiment is already complete.

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Bottom line

The earlier idea—an action-conditioned, partly neural, partly structured 3D/4D memory—is directionally right but too broad for one paper. The 2026 literature makes the narrower opportunity clearer. Keep a persistent object/event world memory outside the transformer, make retrieval into the transformer's bounded working context the research object, and evaluate the accuracy–latency–memory trade-off under dynamic object changes. This creates a credible bridge between robotic intelligence and transformer architecture without committing to an end-to-end robot system.